# *KiCAD GitHub Guide*

**University of California, Davis**

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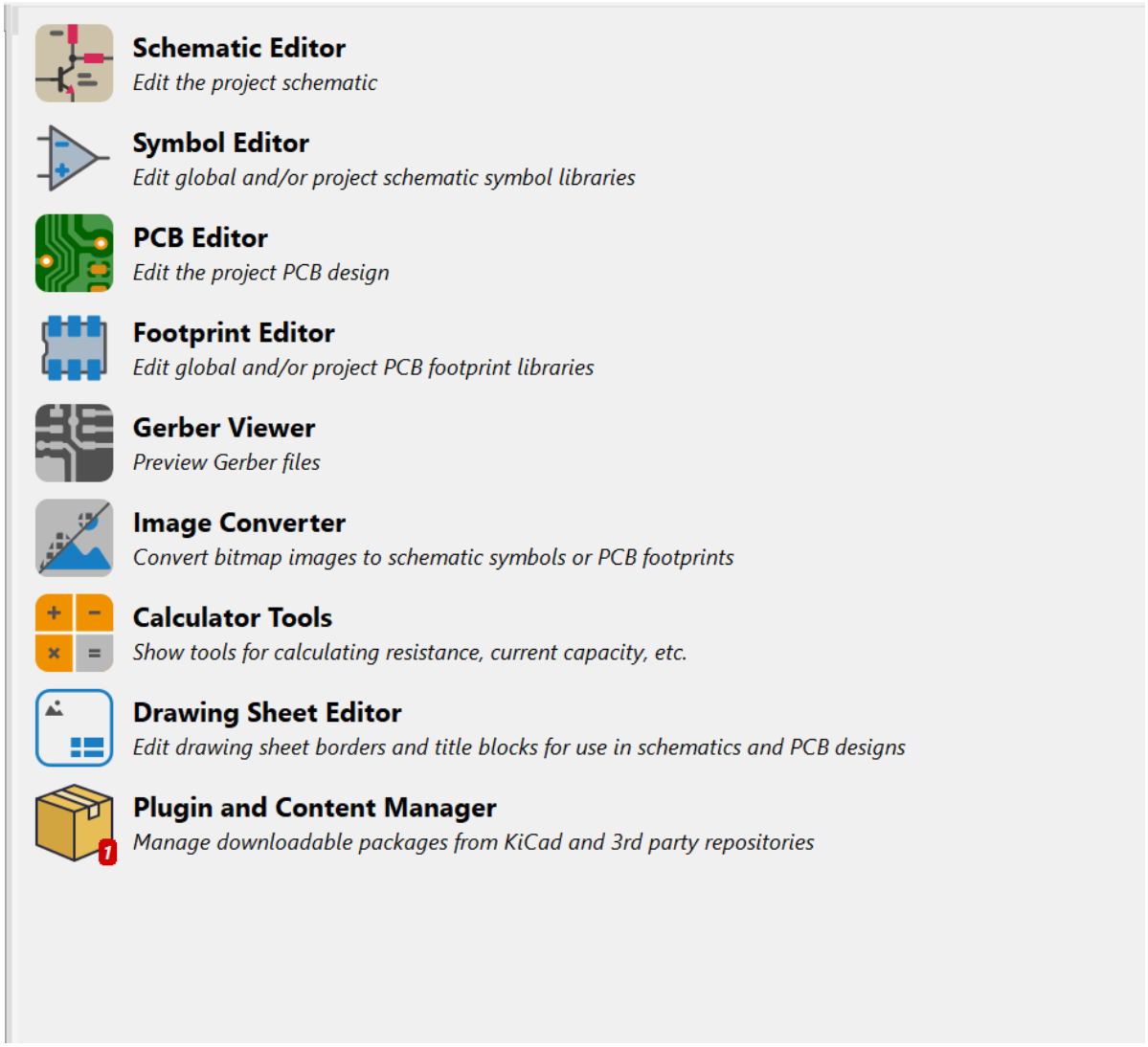
Disclaimer: How we use GitHub for KiCad will differ from how many organizations use GitHub for code. Please read the documentation thoroughly before contributing or making changes.

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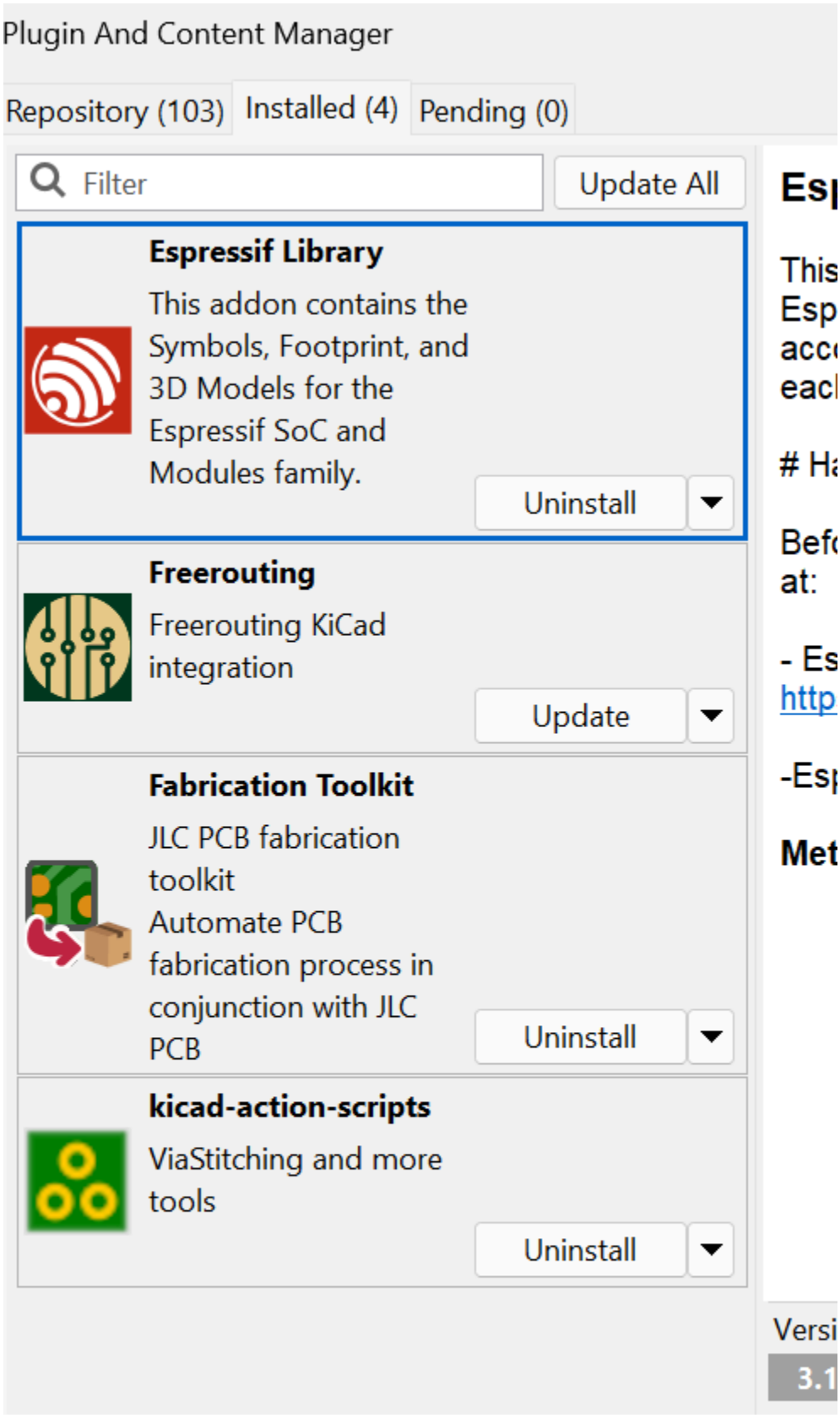
# Account Creation + Downloads

## KiCAD and KiCAD Plugin Download:

1) Download the latest version of KiCAD according to your operating system. The current version of KiCAD is 10.0.1
    a) Download Link: https://www.kicad.org/download/
2) Optional but highly recommended : After installation, open KiCAD and navigate to the *Plugin and Content Manager* tab.



3) Search for and install the following plugins in the *Plugin and Content Manager* tab

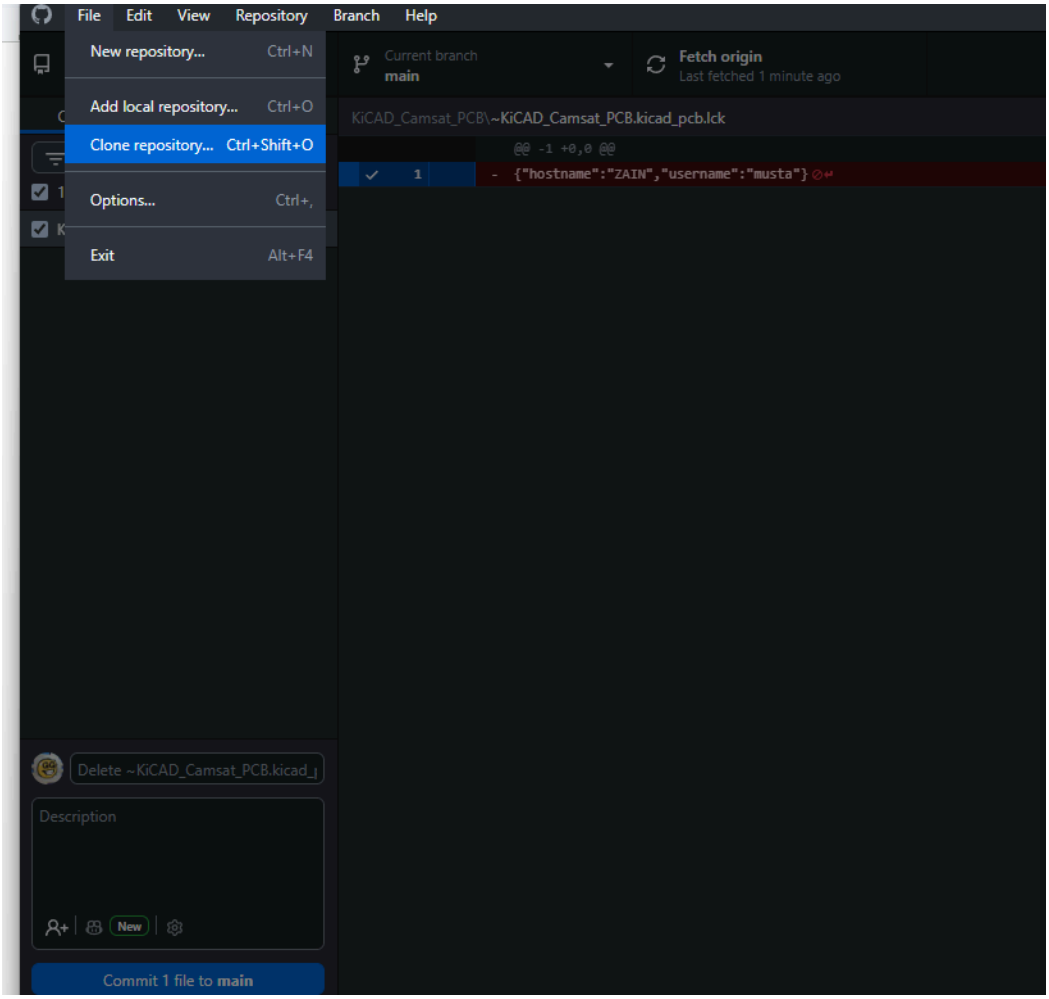


## GitHub Account & Download:

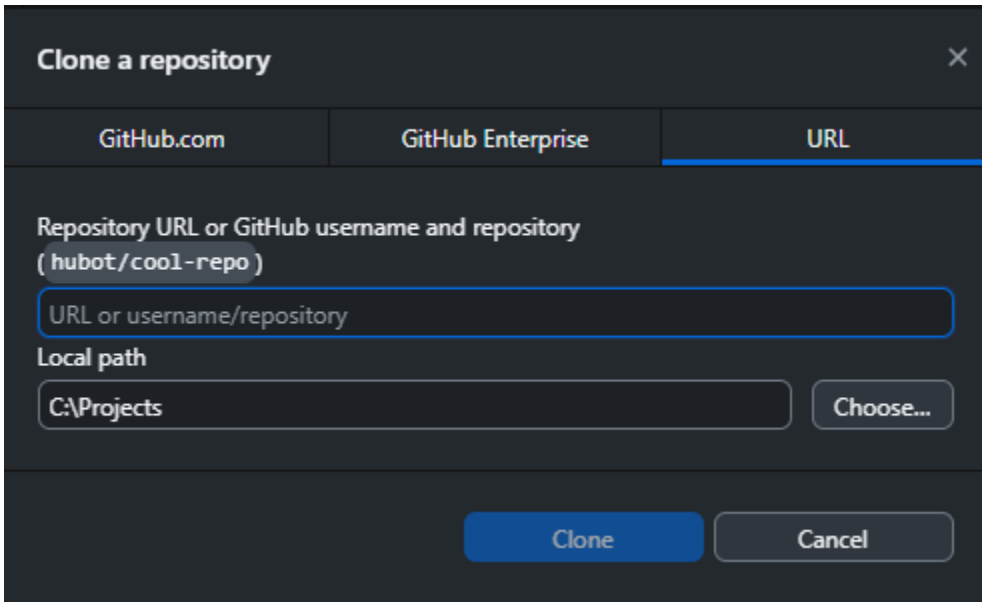
1) Create an account on github using your **PERSONAL** email address: https://github.com/signup
2) Download the latest version of git: https://git-scm.com/
3) Download the latest version of github desktop: https://desktop.github.com/download/
    a) If you know to use git within the terminal this is not necessary but does it life easier
4) Request permission to join the NAR organization on github in order to gain edit access
    a) https://github.com/National-Association-of-Rocketry-UCD

# Project Setup

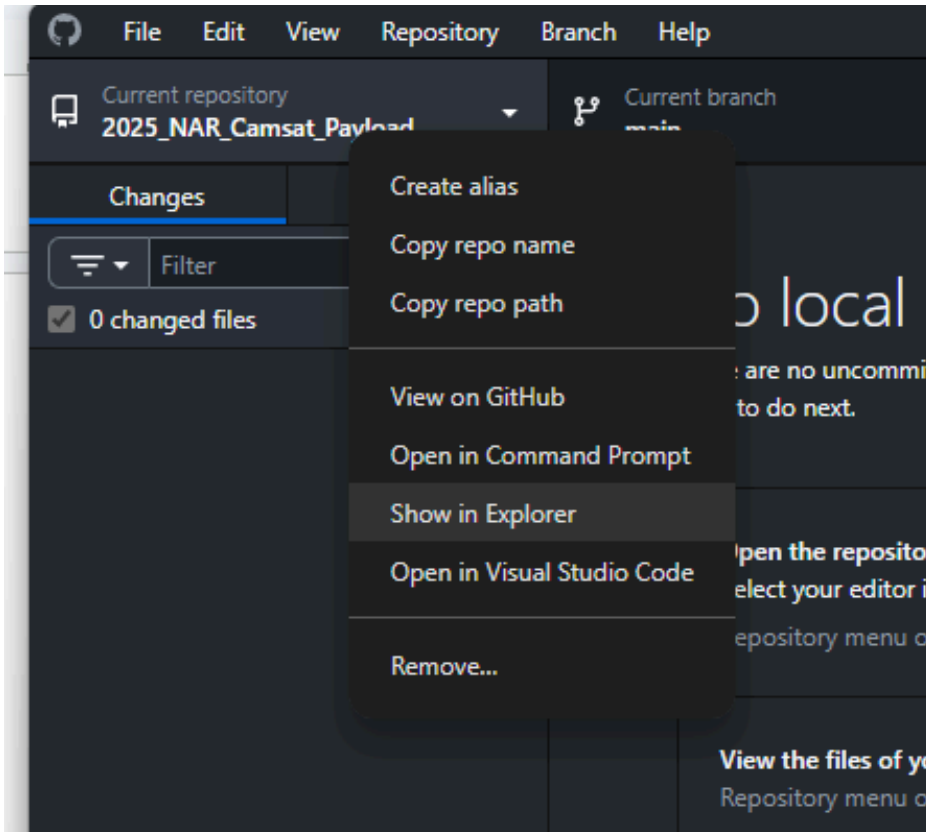
1) Only **AFTER** all software has been downloaded move on to this section
2) Launch Github desktop and navigate to file -> clone repository



3) Navigate to the URL section and paste the github url of the repository for the project which you are currently working on



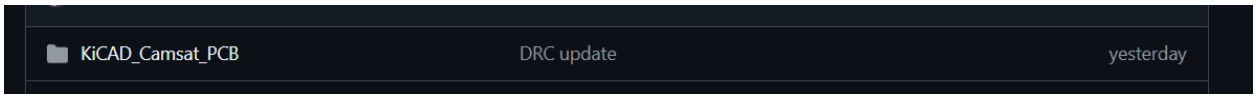

4) To quickly access the files which you have cloned you can right click on the project and press *show in explorer*



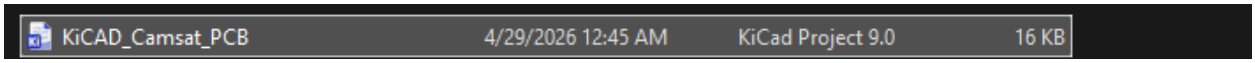
# Working in KiCAD

## Opening the Project

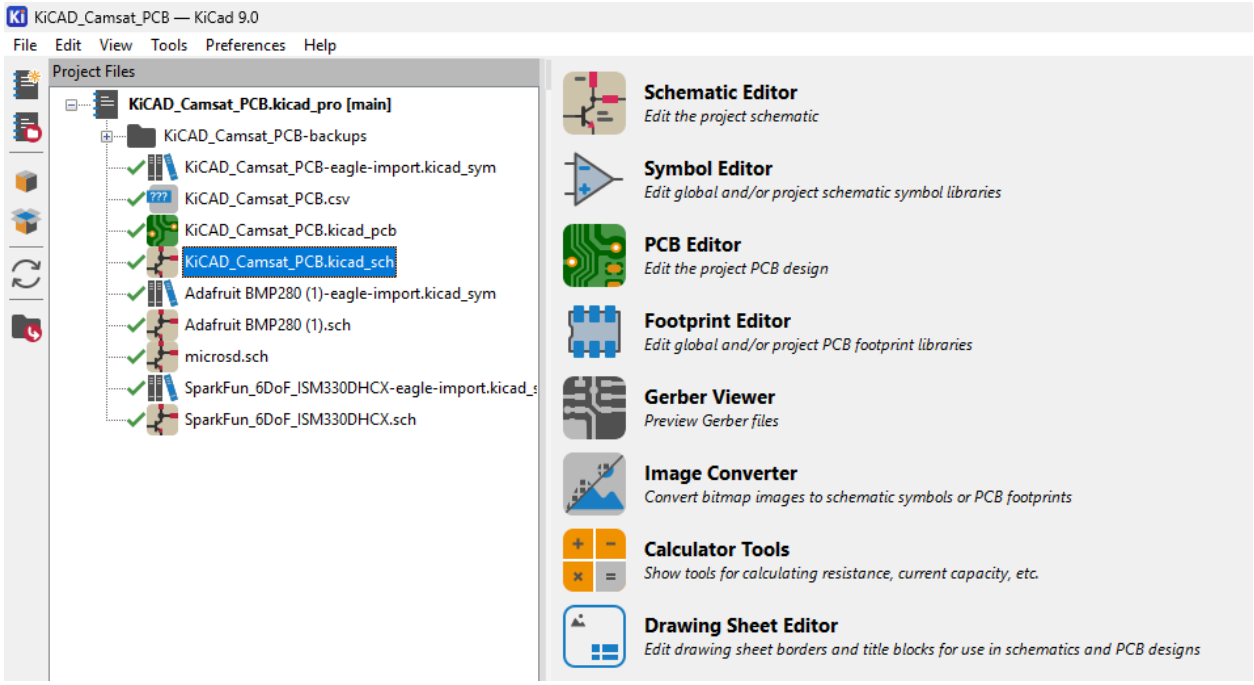
1) Each KiCad project folder in GitHub will be labeled with the prefix **“KiCAD_”** followed by the project title. Navigate to this folder to find the project file



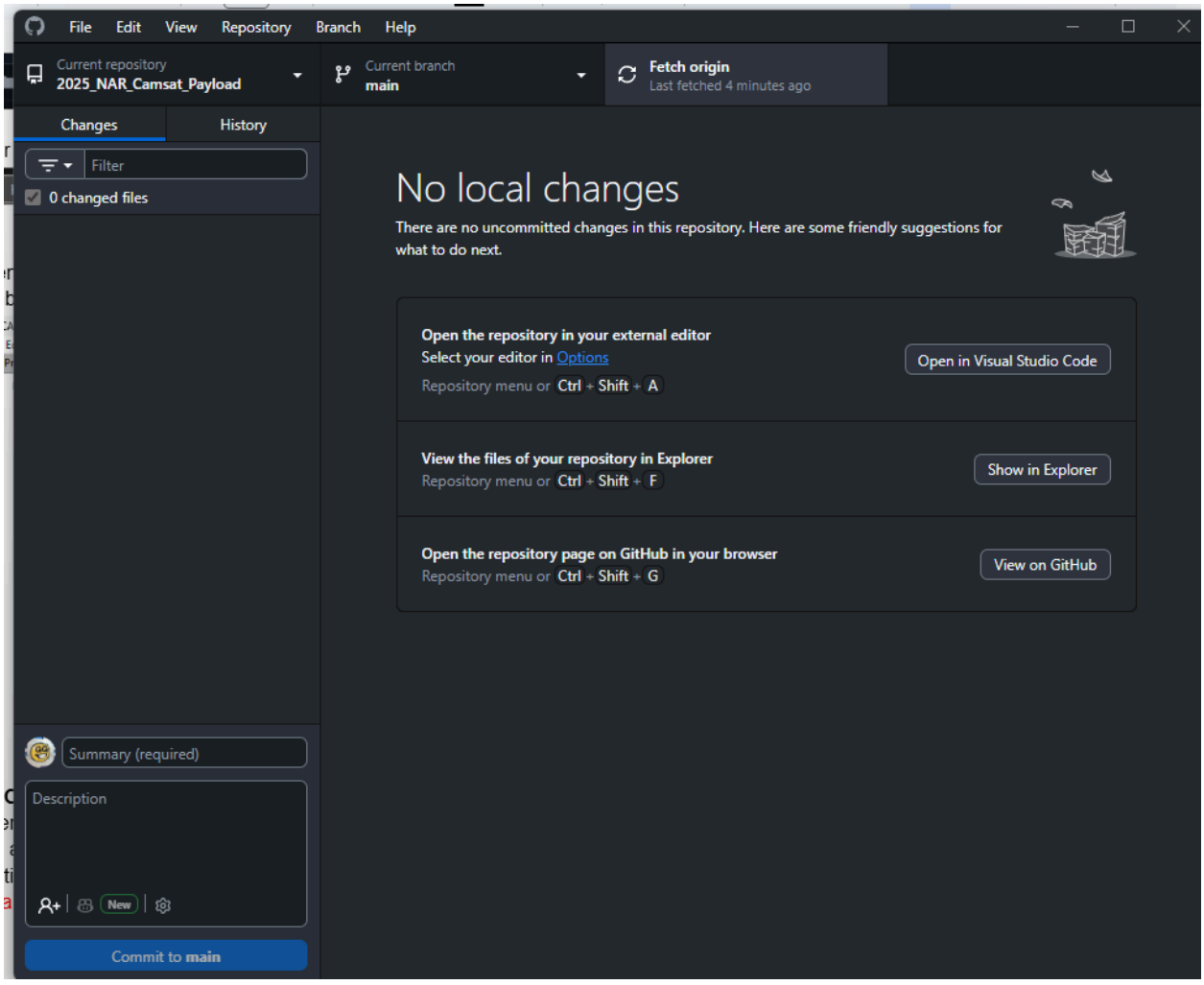

2) After navigating to this folder you can open the project file (.kicad_pro).




3) Opening the project file will open all associated aspects of the project (schematics, symbols, pcb files etc.)



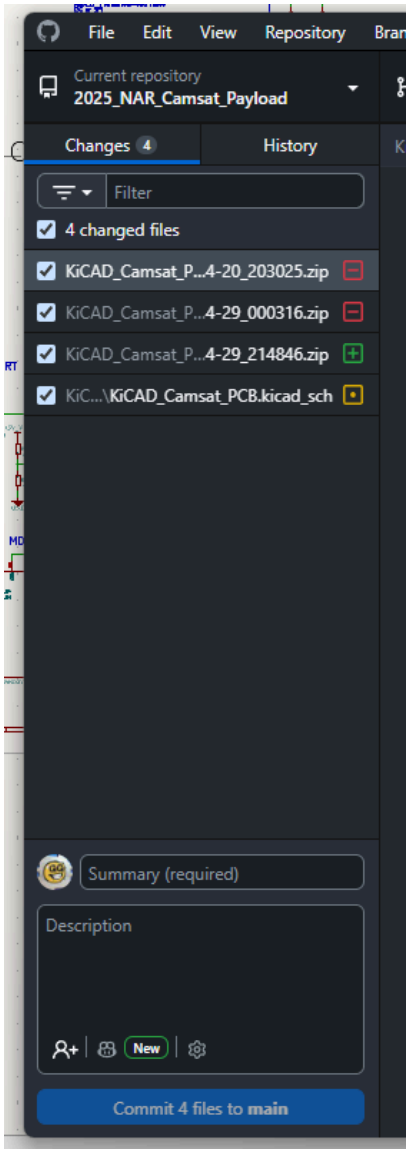
4) **IMPORTANT:** Each time before you start working on the project, click *Fetch Origin*. This updates your design with any changes that other people have made and uploaded to Github.




5) When making changes to the design, you can either create a copy of the schematic/PCB files within the project and make major edits there, or edit the files directly. If creating a copy of an existing file, add your name to the prefix of the new file.
    a) Avoid having two people work on the same parts of the same file simultaneously, as this will lead to merge conflicts. These can be resolved it just requires manual intervention to make sure the proper design changes are kept

## Uploading Changes

1) Once you are ready to save your changes, save the files within KiCAD. Then navigate back to github desktop. It will display the changes to all of the files which you have made.




2) Include a detailed description of the changes which you made and what files you edited then press commit to main.

3) Then click push origin to upload your changes to the repository

